Chapter 6, Problem 3P

Problem

Design a problem to help other students better understand how capacitors work.

Step-by-step solution

Step 1 of 4

Determine the voltage across a $9\text{-}\mu\text{F}$ capacitor, the current through the capacitor is $i(t) = 9e^{-\frac{t}{9000}}\ \text{mA}$. The initial capacitor voltage prior to application of input is, $3\ \text{V}$.

Step 2 of 4

The expression for the voltage across the capacitor is,

$$\begin{aligned} v(t) &= \frac{1}{C}\int_{-\infty}^{t} i(t)\,dt \\ &= \frac{1}{C}\left(\int_{-\infty}^{0} i(t)\,dt + \int_{0}^{t} i(t)\,dt\right) \end{aligned}$$

The initial capacitor voltage is the voltage across the capacitor prior to the application of the input.

The voltage across the capacitor cannot change instantaneously; that is, the voltage across the capacitor must be continuous, as an infinite amount of current is needed to obtain an instantaneous change in the voltage across the capacitor.

The input signal is zero for all time instants less than zero.

The initial voltage is the voltage across the capacitor from time instants, $-\infty$ to 0.

Thus, the initial voltage across the capacitor is, $v(0)$.

Thus, the resultant expression for the voltage across the capacitor is,

$$v(t) = \frac{1}{C}\left(v(0) + \int_{0}^{t} i(t)\,dt\right) \quad \text{...... (1)}$$

Step 3 of 4

Determine the voltage across the capacitor.

Substitute $9e^{-\frac{t}{9000}}\ \text{mA}$ for $i(t)$ and $3\ \text{V}$ for $v(0)$ in equation (1).

$$\begin{aligned} v(t) &= \frac{1}{9\times10^{-6}}\left(3\ \text{V} + \int_{0}^{t} 9e^{-\frac{t}{9000}}\ \text{mA}\ dt\right) \\ &= \frac{1}{9\times10^{-6}}\left(3\ \text{V} + \left|\frac{9e^{-\frac{t}{9000}}\times10^{-3}}{\frac{-1}{9000}}\right|_{0}^{t}\right) \qquad \left(\text{Since} \int e^{-ax} = \frac{e^{-ax}}{-a}\right) \\ &= \frac{1}{9\times10^{-6}}\left(3\ \text{V} + \left|-9e^{-\frac{t}{9000}}\times10^{-3}\times9000\right|_{0}^{t}\right) \\ &= \frac{1}{9\times10^{-6}}\left(3\ \text{V} + \left|-81e^{-\frac{t}{9000}}\right|_{0}^{t}\right) \end{aligned}$$

Further simplification gives:

$$\begin{aligned}v(t) &= \frac{1}{9\times10^{-6}}\left(3\text{ V} + -81e^{-\frac{t}{9000}} + 81e^{-\frac{0}{9000}}\right)\\ &= \frac{1}{9\times10^{-6}}\left(3\text{ V} + -81e^{-\frac{t}{9000}} + 81e^{-\frac{0}{9000}}\right)\\ &= \frac{1}{9\times10^{-6}}\left(3\text{ V} + -81e^{-\frac{t}{9000}} + 81\right)\\ &= \frac{1}{9\times10^{-6}}\left(84 - 81e^{-\frac{t}{9000}}\right)\end{aligned}$$

Thus, the voltage across the capacitor at time t is,

$$\boxed{v(t) = \frac{1}{9\times10^{-6}}\left(84 - 81e^{-\frac{t}{9000}}\right)}$$